

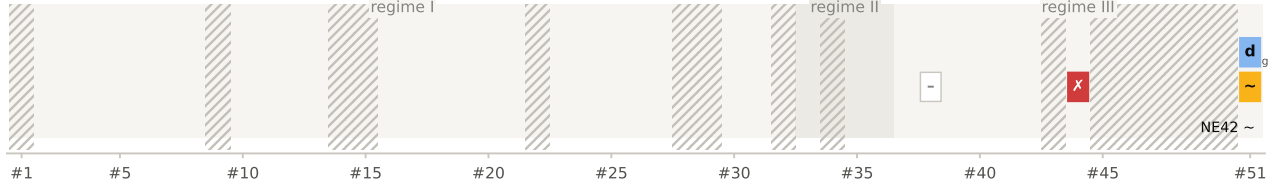
H58: Effective superagents are coordinated agents plus their artifacts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2/5 Usefulness 2.5/5

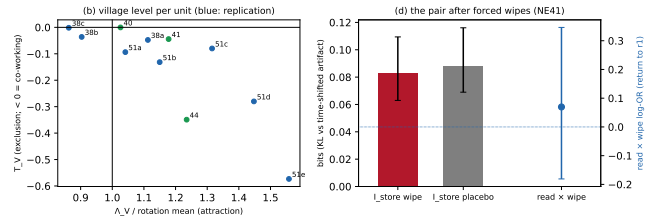
Bottom line. The Kolchinsky–Wolpert agent is the agent with its own artifact; coordination adds small attraction crews, not a superagent.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The claim: coordinated agents plus their artifacts form *effective superagents*. Round 1 found none with a which-artifact state: **agent + own artifact** stayed the unit. Round 2 asks three questions. (R1) Does any unit show *attraction*, with members moving onto each other’s current repo beyond their own histories and beyond outsiders, or *territoriality*, with members keeping off each other’s repo? (R2) Inside shared repos, does a store live at file or issue level? (R3) Is the agent + own-artifact pair the Kolchinsky–Wolpert agent? What semantic information does it hold after a forced context wipe? **Result:** the pair holds its next-file information through the wipe, and only its own files carry identified semantic information. Small crews in #51b–c add a specific attraction term; it is few joins. Territoriality is not found as a coordinated state.



Observables. Left: one point per unit. Villages co-work on shared repos ($T_V < 0$ in 11/11 defined units) and lean to attraction ($A_V > 1$ in 9/11). None shows exclusion. Right: after 10,806 forced wipes the pair’s next-file information against a time-shifted artifact is 0.083 bits, against 0.088 at placebo calls. Reading the last-committed file predicts a return to it equally in both arms (interaction +0.07).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Treat one agent and its own repository as the unit. Its “which file next” information lives in the repo and survives a context wipe (0.94 of placebo). Re-reading its own files is worth about 0.8 work commits per bit; re-reading other agents’ files or goal documents is worth nothing measurable. Keep each agent’s work state in its own files. Do not expect a wipe to send agents to the group’s files. Small crews do move together onto each other’s repos (#51c crew of six: 21 joins vs 3.0 expected). The effect is a small fraction of moves and may be chat-steered.

PHYSICAL MODEL

Semantic information (model 04). Member j moves from its own last repo ℓ_j to k ; O_q is the set of repos agent q occupies in bins $b, b + 1$. π_j is j ’s leave-one-day-out repo popularity. Attraction Λ compares joins with that expectation. Exclusion T compares same-repo pair-bins with within-day rotations. In R3, X is the next commit’s file code and S the first re-read file code; $\tilde{X}$ codes X against the agent’s artifact state on another day.

$$\Lambda_G = \frac{\sum_{j \neq q \in G} \#\{k \in O_q\}}{\sum_{j \neq q \in G} \sum_{o \in O_q} \pi_j(o) / [1 - \pi_j(\ell_j)]}$$

$$T_G = 1 - C_{\text{obs}} / \langle C_{\text{rot}} \rangle$$

$$I_{\text{store}} = D_{\text{KL}}(p(X) \| p(\tilde{X})), \quad \kappa = \Delta V / I(X; S)$$

RESULTS VS PREDICTIONS

prediction	observed	verdict
R1a no attraction	51b 2, 51c 6, 51d 1 pass; crew	failed
unit in #51b-d	of 6 robust to bin width	
R1b village herding	both tests in 2/12; $T_V < 0$ in not supp.	
$\geq 2/3$	11/11	
R1c territorial group	0 of 331; power ≤ 0.5	inconcl.
R1d #51e trio terri-	never shares a repo on any day	failed
torial		
R2a own file in	31/46 repos (non-specific)	supported
shared repo		
R2b no file-level	pooled z 5.4; 1.9 without one	failed*
store	repo	
R2c file exclusion	pooled z -6.4; co-editing	failed
R3a store survives	ratio 0.94 [0.79, 1.10]; 0.083 bits	mixed
wipe	(<0.1)	
R3b re-read re-	+0.07 [-0.18, +0.35]; power 1.0	failed
acquires		
R3c in-flight $\Delta I \ni 0$	+0.009 [-0.018, +0.033] bits	supported
R3d κ own > others,	own I 0.19 bits, κ 0.77; others	supported
goal	0.001	

*by the pooled letter; the per-repo rule holds (35/39 below).

GOAL PERIODS IN DETAIL

G44 × failed
G51 ~ mixed
G51g d descriptive
NE42 ~ mixed
G38 - n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	2		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Crews and search sets were defined on the same data; the search sets are selection-inflated. The attraction joins are few (5–35 per unit). For the #51c crew of six the rotation z falls to 1.7 once human-message bins are removed. The #44 and #51e qualifiers coincide with human messages. Joins were not read-gated, so chat steering (H11) is not excluded. The territorial rule is unpowered. The file-level positive rests on one repository. Issue level covers one repository. Reads are parsed command tokens. Commits are attempts. The reserved data (#51 tail, NE24) are not used.

ROUND 2

Read-gate the crew joins (read vs posted-but-unread). Define crews on half the days and test on the other half. Measure whether joins raise output. Run the re-frozen `confirm.py` on the #51 tail.

- **Round-3 redirects (newest).**
- **H58-R5** Read-gate the joins. For the #51b–c attraction crews: does a join follow a read of the co-member’s message or commit (context ledger) more than a posted-but-unread one at a matched lag?
- **H58-R6** Out-of-sample crews. Define crews on the first half of a unit’s days and test the A-rule on the second half (removes in-sample selection).
- **H58-R7** Value of the crew. Do attraction-crew members produce more (work commits, or outcome-resolved writes) in bins after a join than after a matched non-join move?
- **H58-R4** (carried). Run the powered reserved data (#51 tail) through a re-frozen `confirm.py` that adds the A-rule and the R3 store and read × wipe statistics.

Second figure. Left: synthetic validation on each unit’s real activity figure. The attraction rule finds a planted group store with power ≥ 0.85 in #51b–d already at $\rho = 0.35$, with size ≤ 0.05 under own persistence and a common field. The territorial rule reaches only 0.5. Right: file-level statistics per shared repo; exclusivity is high but independent owners produce it too.

